Network Forensics Examination Report

OWASP Juice Shop traffic analysis using Parrot OS and Wireshark

Investigator: Drew

Examination date: November 4, 2026

Capture system: Parrot OS virtual machine

Target application: OWASP Juice Shop hosted on the Mac system

1. Executive Summary

This examination documented network communication between a Parrot OS virtual machine and an OWASP Juice Shop web application hosted on a Mac system. Wireshark was used on Parrot OS to capture HTTP traffic on the enp0s1 interface. Three HTTP GET requests and their corresponding responses were identified: a homepage request, a product-search API request, and a request to an unrecognized test path. All three responses were reported as HTTP 200 OK.

The packet capture was supplemented with Parrot OS system journal records, network interface and routing information, and shell history. These sources support identification of the client address, network interface, capture activity, and commands associated with the three observed requests. Firefox history was examined as a potential source but was not expected to contain the activity because the requests were generated with curl.

The examination is limited to the traffic and artifacts actually collected. The results do not establish malicious activity or prove a vulnerability. The HTTP 200 response for /forensic-test does not prove that the route is a valid application resource; it may reflect a single-page-application fallback.

2. Scope, Objective, and Environment

2.1 Objective

The objective was to capture and analyze network traffic generated by accessing OWASP Juice Shop, examine relevant operating-system and network artifacts, construct an event timeline, and correlate independent evidence sources.

2.2 Scope and boundaries

- Target application: OWASP Juice Shop in an authorized lab environment.
- Client: Parrot OS virtual machine.
- Capture tool: Wireshark.
- Traffic-generation utility: curl.
- The saved capture documents HTTP traffic between 192.168.64.3 and 192.168.64.1 on TCP port 3000.
- The Python penetration-testing toolkit was not used to generate the traffic analyzed in this examination.

2.3 Network environment

Item	Observed value	Significance
Client system	Parrot OS virtual machine	Origin of the captured HTTP requests
Client interface	enp0s1	Wireshark capture interface
Client IPv4 address	192.168.64.3/24 (dynamic)	Source address in captured requests
Client MAC address	52:54:00:27:43:47	Interface identifier; consistent with a virtual NIC
Server/host address	192.168.64.1	Destination address hosting Juice Shop
Destination TCP port	3000	Published Juice Shop service port
Time standard	UTC (+00:00)	Used for journal and capture-time correlation

3. Methodology and Tools

The examination used the following workflow:

1. Start Wireshark on Parrot OS and capture traffic on enp0s1.
2. Generate HTTP requests from Parrot OS using curl to the Juice Shop host at 192.168.64.1:3000.
3. Apply Wireshark display filters to isolate HTTP requests and responses.

4. Inspect request and response details, including endpoints, methods, content types, and status codes.
5. Save the packet capture in PCAPNG format.
6. Collect Parrot OS system journal, kernel/network events, interface configuration, and routing information.
7. Examine TCP socket state and shell history as supplementary artifacts.
8. Identify the Firefox history database and assess its relevance to the activity.
9. Compare timestamps and network identifiers across the available artifacts.

4. Packet Capture and HTTP Analysis

4.1 Capture configuration

Property	Value
Capture application	Wireshark
Operating system	Parrot OS
Interface	enp0s1
Capture file	juice_shop_network_capture.pcapng
Protocol observed	HTTP over TCP
Client	192.168.64.3
Server	192.168.64.1
Server port	3000

The capture was inspected using the Wireshark display filter `http`. The filter exposed the three request/response pairs listed below. The `http.request` and `http.response` display filters can be used to isolate requests and responses separately.

4.2 Observed HTTP exchanges

Packet No.	Relative time (s)	Direction	Observed event
5	0.001290249	192.168.64.3 → 192.168.64.1	GET / HTTP/1.1
7	0.004911328	192.168.64.1 → 192.168.64.3	200 OK (text/html)
20	6.871813246	192.168.64.3 → 192.168.64.1	GET /rest/products/search?q=apple HTTP/1.1
23	6.885092607	192.168.64.1 → 192.168.64.3	200 OK (application/json)
35	14.902631395	192.168.64.3 → 192.168.64.1	GET /forensic-test HTTP/1.1
38	14.913505926	192.168.64.1 → 192.168.64.3	200 OK (text/html)

4.3 HTTP response and header observations

An earlier Follow TCP Stream inspection of the homepage exchange showed an HTTP 200 OK response, a text/html content type, and a Content-Length of 9,393 bytes. The response headers observed in that inspection included:

Header	Observed value	General purpose
X-Content-Type-Options	nosniff	Reduces MIME-type sniffing behavior.
X-Frame-Options	SAMEORIGIN	Restricts framing to the same origin in supporting browsers.
Access-Control-Allow-Origin	*	Allows cross-origin access; security impact depends on the resource and context.

Feature-Policy	payment 'self'	Legacy policy header restricting a browser feature.

The wildcard Access-Control-Allow-Origin value is an observation, not by itself proof of a vulnerability. Its significance depends on the resources exposed and the application's security design.

4.4 Unrecognized route response

The request to /forensic-test returned HTTP 200 OK with a text/html response. The response length was reported as matching the homepage response in the packet-list comparison. This behavior may be consistent with a single-page application returning its main HTML document for unknown paths and allowing client-side JavaScript to handle routing. The observed 200 status must not be reported as proof that /forensic-test is a valid application page. Compare HTTP response-body content, not only the Wireshark frame Length column, before asserting that two bodies are identical.

5. System and Network Log Examination

5.1 Collected files

File	Reported size	Purpose
system_journal.log	957 KB	System journal entries for the collection day
network_events.log	2.4 KB	Filtered kernel/network-related events
network_interfaces.txt	891 bytes	Network interface addresses and state
routing_table.txt	156 bytes	Routing configuration

The files were collected under `~/network_forensics/logs/` using journalctl and ip commands.

5.2 Time reference

The Parrot OS date command returned `2026-11-04T17:31:04+00:00`. The virtual machine therefore reported UTC (+00:00) at collection time. Journal timestamps and Wireshark absolute timestamps should be compared using this time standard. Wireshark relative timestamps are offsets from the beginning of a capture and are not calendar timestamps.

5.3 Kernel and interface events

The network event log contained a boot-time message indicating that virtio_net renamed eth0 to enp0s1. This supports identification of enp0s1 as the virtual network interface used for capture. Unrelated ACPI interrupt-link messages matched by the broad text filter were treated as noise.

Promiscuous mode entered (UTC)	Promiscuous mode left (UTC)	Duration
16:58:36	17:00:21	1 min 45 sec
17:00:38	17:00:59	21 sec
17:02:40	17:03:18	38 sec
17:06:49	17:07:17	28 sec
17:07:28	17:07:37	9 sec
17:07:43	17:08:51	68 sec
17:09:52	17:10:19	27 sec
17:11:44	17:12:02	18 sec
17:18:22	17:19:10	48 sec
17:23:34	17:23:59	25 sec

These entries record periods when the interface entered and left promiscuous mode, consistent with Wireshark capture activity. They corroborate capture activity but do not, by themselves, identify the packets contained in a particular saved capture.

5.4 DHCP and address assignment

At 17:16:51 UTC, NetworkManager recorded a DHCP lease event for enp0s1 with address 192.168.64.3. The dispatcher service started following the lease change and deactivated normally at approximately 17:17:01 UTC. The DHCP event occurred between the 17:11:44–17:12:02 and 17:18:22–17:19:10 promiscuous-mode sessions. It supports the attribution of 192.168.64.3 to the Parrot interface during the later capture sessions.

The reported Unix epoch value in the journal was consistent with 2026-11-04 17:16:51 UTC. The interface snapshot showed enp0s1 UP with dynamic IPv4 address 192.168.64.3/24 and MAC address 52:54:00:27:43:47. The remaining lease duration was noted as 2,719 seconds, but the exact lease duration and renewal behavior were not independently confirmed; therefore, the remaining-time value is not used to assert an exact snapshot timestamp.

6. Connection-State Examination

The command `sudo ss -antp` was run to inspect TCP sockets and associated processes. The reported output contained only the column header and no socket rows.

This is a point-in-time negative result: no TCP connections or listening sockets were listed when the command was executed. It is consistent with short-lived curl connections having closed after their requests completed. It does not establish that no earlier connections occurred.

A journal search for curl, socat, Wireshark, and network-related terms returned a NetworkManager DHCP event but no entries identifying those applications. This is not evidence that the applications were not used; their activity was primarily visible in terminal output and the packet capture rather than the system journal.

7. Browser History Examination

A Firefox history database was identified at `/home/user/.mozilla/firefox/wobaigx.default-esr/places.sqlite`. A copy was prepared for examination under `~/network_forensics/browser/places.sqlite`.

The Juice Shop requests in this examination were generated using curl from Parrot OS rather than through Firefox. Therefore, Firefox history was not expected to contain the three HTTP requests. This is documented as a limitation and not as evidence that the network activity did not occur.

8. Shell History Examination

The Parrot shell history contained the three most recent relevant curl commands, corresponding in order to the three GET requests observed in the saved capture:

- GET request to http://192.168.64.1:3000/
- GET request to http://192.168.64.1:3000/rest/products/search?q=apple
- GET request to http://192.168.64.1:3000/forensic-test

The history also contained earlier curl variants, including HEAD requests, SQL-injection-style search probes, an open-redirect URL, a request to 127.0.0.1:8080/JSON/..., and a socat relay command involving port 3001. These additional commands were not identified in the saved capture and were not attributed to that capture.

Shell history is supporting evidence of command activity, not conclusive proof that each command executed successfully. The Wireshark packets provide direct evidence of the three observed HTTP exchanges.

9. Correlated Event Timeline

The following table separates relative packet timestamps from absolute system timestamps. Packet times are offsets from the beginning of the saved capture; system events are UTC wall-clock times.

Time	Source	Event	Evidence reference
16:38:24 UTC	Kernel journal	virtio_net interface renamed from eth0 to enp0s1 (boot-time event).	LOG- network_events.log
17:16:51 UTC	NetworkManager journal	DHCP lease assigned to enp0s1: 192.168.64.3.	LOG- system_journal.log
17:17:01 UTC	System journal	NetworkManager dispatcher deactivated normally.	LOG- system_journal.log
17:18:22–17:19:10 UTC	Kernel journal	Promiscuous-mode session consistent with Wireshark capture activity.	LOG- network_events.log
0.001290249 s	Wireshark	GET / requested by 192.168.64.3.	NET-002
0.004911328 s	Wireshark	Homepage response: HTTP 200 OK, text/html.	NET-002
6.871813246 s	Wireshark	GET /rest/products/search?q=apple requested.	NET-003
6.885092607 s	Wireshark	Search response: HTTP 200 OK, application/json.	NET-003
14.902631395 s	Wireshark	GET /forensic-test requested.	NET-004
14.913505926 s	Wireshark	Response: HTTP 200 OK, text/html.	NET-004
17:23:34–17:23:59 UTC	Kernel journal	Promiscuous-mode session consistent with a later Wireshark capture.	LOG- network_events.log
17:31:04 UTC	System clock	Collection time reference recorded as UTC.	Collection notes

The exact absolute start and end times for the saved PCAPNG file should be retained from the capinfos output and added to the final evidence register. The packet-relative timestamps above must not be directly compared to journal wall-clock timestamps without that conversion.

10. Findings and Interpretation

- The saved Wireshark capture contains three HTTP GET requests from 192.168.64.3 to 192.168.64.1 on TCP port 3000 and three corresponding server responses.
- The responses were reported as HTTP 200 OK: HTML for the homepage, JSON for the product-search endpoint, and HTML for /forensic-test.
- NetworkManager and interface configuration records support the attribution of 192.168.64.3 to Parrot OS interface enp0s1.
- Kernel promiscuous-mode records corroborate Wireshark capture activity on enp0s1.
- Shell history contains commands matching the three requests in the saved capture.
- No active TCP sockets were listed at the later ss collection time; this does not negate earlier completed connections.
- Firefox history was not expected to contain the requests because curl generated the activity.
- The /forensic-test HTTP 200 response may reflect application fallback behavior and does not prove the route exists.

11. Evidence Inventory and Preservation

The following artifacts were identified during the examination. Replace the provisional references below with the actual unique evidence IDs from the project's evidence register if different.

Provisional ID	Artifact	Description / preservation note
NET-001	juice_shop_network_capture.pcapng	Original Wireshark packet capture; retain unchanged.
LOG-001	system_journal.log	Parrot OS system journal export.
LOG-002	network_events.log	Filtered kernel/network event export.
LOG-003	network_interfaces.txt	Interface state, IP and MAC information.
LOG-004	routing_table.txt	Routing table snapshot.
BR-001	places.sqlite	Copy of Firefox history database; relevance limited because curl generated test traffic.
SH-001	Shell history output	Relevant command history; preserve the source history file or an appropriately collected copy.
INT-001	SHA256SUMS.txt	Hash manifest for collected files, once generated and verified.

Hash verification status: pending confirmation of the final SHA-256 output. Do not mark the evidence integrity check as complete until the hashes have been generated, recorded, and independently rechecked.

Collection handler: Drew. Collection timestamps and timezone: record the actual collection time for each artifact; the VM time reference observed during collection was UTC (+00:00). Capture first/last absolute timestamps: insert the exact values from capinfos.

12. Limitations

- The examination used a limited set of HTTP requests and does not represent a comprehensive capture of all Juice Shop traffic.
- The browser-history source was not expected to contain curl-generated requests.
- The ss command records current socket state and cannot reconstruct closed historical connections.
- The system journal did not independently log the curl, socat, or Wireshark commands.
- Shell history can be incomplete and does not independently prove successful execution.
- The saved capture's relative timestamps require absolute capture start/end times for exact correlation with journal events.
- The observed HTTP 200 response for /forensic-test does not establish that the route is a valid application resource.

13. Conclusion

The examination successfully captured and analyzed HTTP traffic between Parrot OS and OWASP Juice Shop. Wireshark provided direct evidence of three HTTP request/response exchanges. Parrot OS system logs, DHCP records, interface configuration, and shell history provided supporting context and corroboration. The artifacts were used to construct a sequence of network events and identify the limitations of each evidence source.

The results support the conclusion that Parrot OS at 192.168.64.3 communicated with the Juice Shop host at 192.168.64.1 over HTTP on TCP port 3000 during the captured activity. No conclusion of compromise or exploitation is made from this network capture alone.

Appendix A — Commands Used

```
date -Ins
sudo journalctl --since "today" --no-pager > ~/network_forensics/logs/system_journal.log
sudo journalctl -k --since "today" --no-pager | grep -Ei 'enp0s1|network|link|dhcp|ethernet' >
~/network_forensics/logs/network_events.log
ip addr show > ~/network_forensics/logs/network_interfaces.txt
ip route show > ~/network_forensics/logs/routing_table.txt
sudo ss -antp
sudo journalctl --since "2026-11-04 17:00:00" --until "2026-11-04 18:00:00" --no-pager | grep -Ei
'socat|curl|wireshark|network'
cat ~/.bash_history | grep -Ei 'curl|socat|wireshark'
find ~/.mozilla/firefox -name places.sqlite
capinfos juice_shop_network_capture.pcapng
sha256sum juice_shop_network_capture.pcapng
```

Log Parser Analysis

1. Purpose

Microsoft Log Parser 2.2 was selected as the log-analysis tool for examining system and network-related log data. The tool provides SQL-style querying capabilities for filtering, sorting, grouping, and summarizing structured log information. It supports data sources such as CSV, XML, text files, and Windows Event Logs.

The purpose of this analysis was to identify relevant network events, examine activity during the forensic investigation period, and correlate log entries with the packet capture and other collected evidence.

2. Tool Information

- **Tool:** Microsoft Log Parser 2.2

- **Version:** 2.2.10

- **Developer:** Microsoft

- **Primary function:** Structured log analysis using SQL-style queries

- **Intended input:** Exported system journal and network event data

- **Intended output:** Filtered records and summarized results

3. Evidence Sources

The analysis was designed around the following evidence collected from the Parrot OS forensic environment:

- `system_journal.log` — system journal entries.

- `network_events.log` — network interface and related system events.

- `network_interfaces.txt` — network interface configuration.

- `routing_table.txt` — network routing information.

- `juice_shop_network_capture.pcapng` — Wireshark packet capture containing HTTP requests and responses.

The system journal was also exported in structured JSON format to support conversion into a format suitable for Log Parser analysis.

4. Analysis Methodology

The intended workflow consisted of:

1. Export system journal records from Parrot OS.

2. Convert the structured journal data into CSV format.

3. Transfer the CSV file to the Windows environment running Log Parser 2.2.

4. Use SQL-style queries to filter and examine relevant records.

5. Identify network-related events and timestamps.

6. Compare relevant timestamps with the Wireshark packet capture.

7. Preserve the resulting query output as supporting forensic evidence.

5. Planned Log Parser Queries

The following queries were prepared as examples of the analysis to be performed against the converted CSV data. The CSV must contain the corresponding column names.

****Query A — Identify network-related events****

```sql

SELECT Timestamp, Message

FROM system_journal.csv

WHERE Message LIKE '%enp0s1%'

```

**Query B — Identify DHCP activity**

```sql

SELECT Timestamp, Message

FROM system_journal.csv

WHERE Message LIKE '%dhcp%'

```

Query C — Identify network interface changes

```sql

SELECT Timestamp, Message

FROM system_journal.csv

WHERE Message LIKE '%link%'

```

Query D — Summarize events by priority

```sql

SELECT Priority, COUNT(*) AS EventCount

FROM system_journal.csv

GROUP BY Priority

ORDER BY EventCount DESC

```

These queries are intended to narrow the dataset to records relevant to network activity and system events.

6. Evidence Correlation
```

The log-analysis workflow was designed to support correlation with the Wireshark capture.

Relevant events included:

- Network interface identification and configuration.

- DHCP address assignment for interface `enp0s1`.

- Network capture activity indicated by interface promiscuous-mode events.

- HTTP requests from `192.168.64.3` to the Juice Shop service at `192.168.64.1:3000`.

The packet capture provides direct evidence of the observed HTTP exchanges, while system logs provide additional context about the network environment and interface activity.

7. Results and Execution Status

Execution status: Pending verification.

The system and network log files were collected from the Parrot OS environment. The Log Parser query workflow and analysis objectives are documented above. Actual execution of Microsoft Log Parser 2.2, query output, and screenshots must be added after the tool is run.

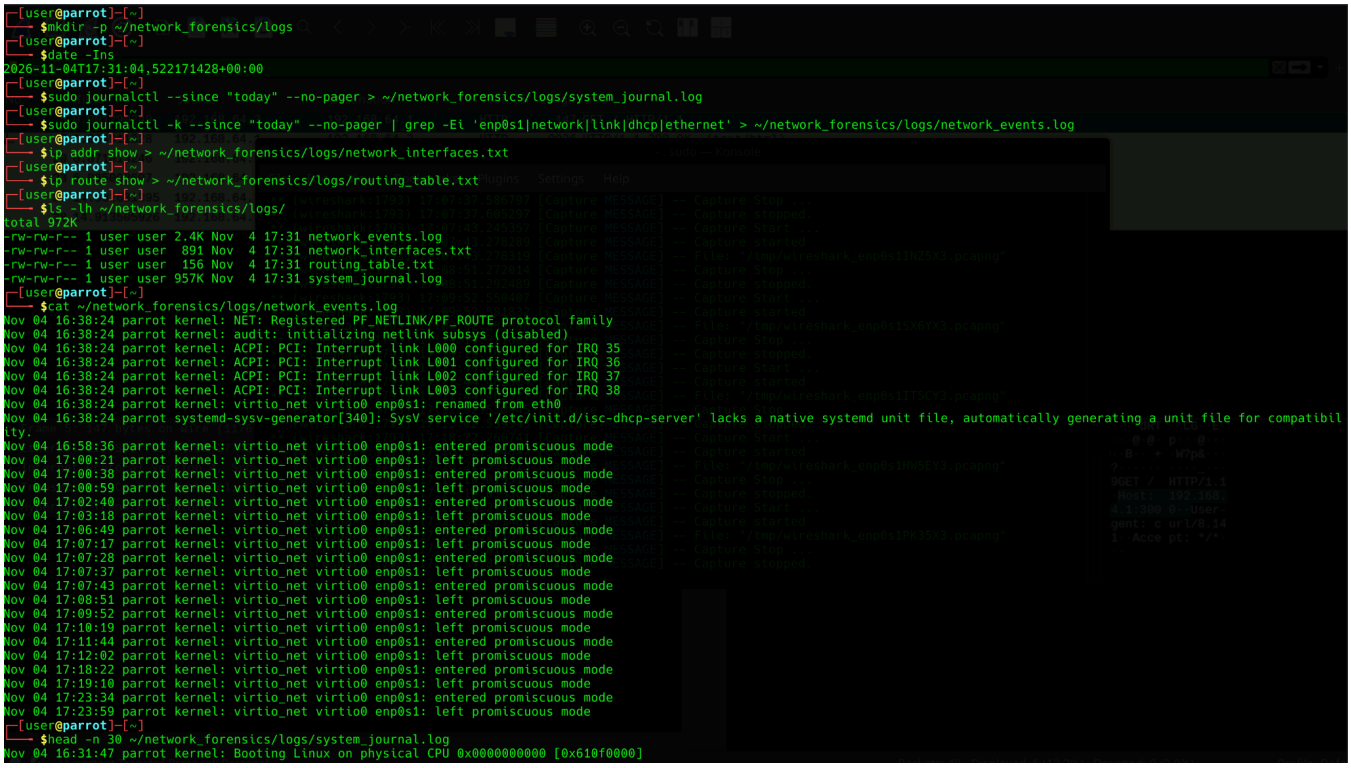
No specific query results or additional findings are attributed to Log Parser until they have been produced and verified.

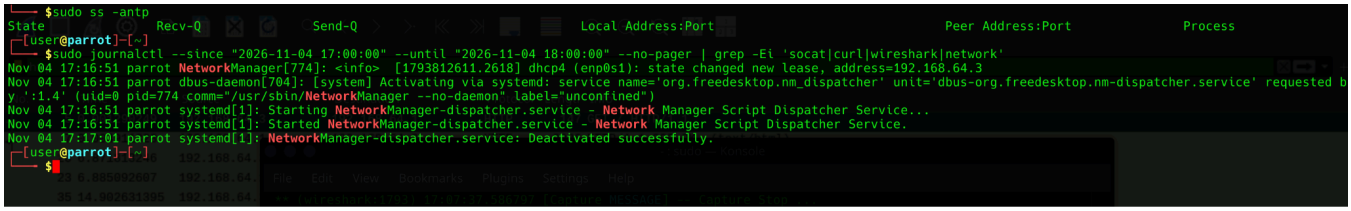
8. Conclusion

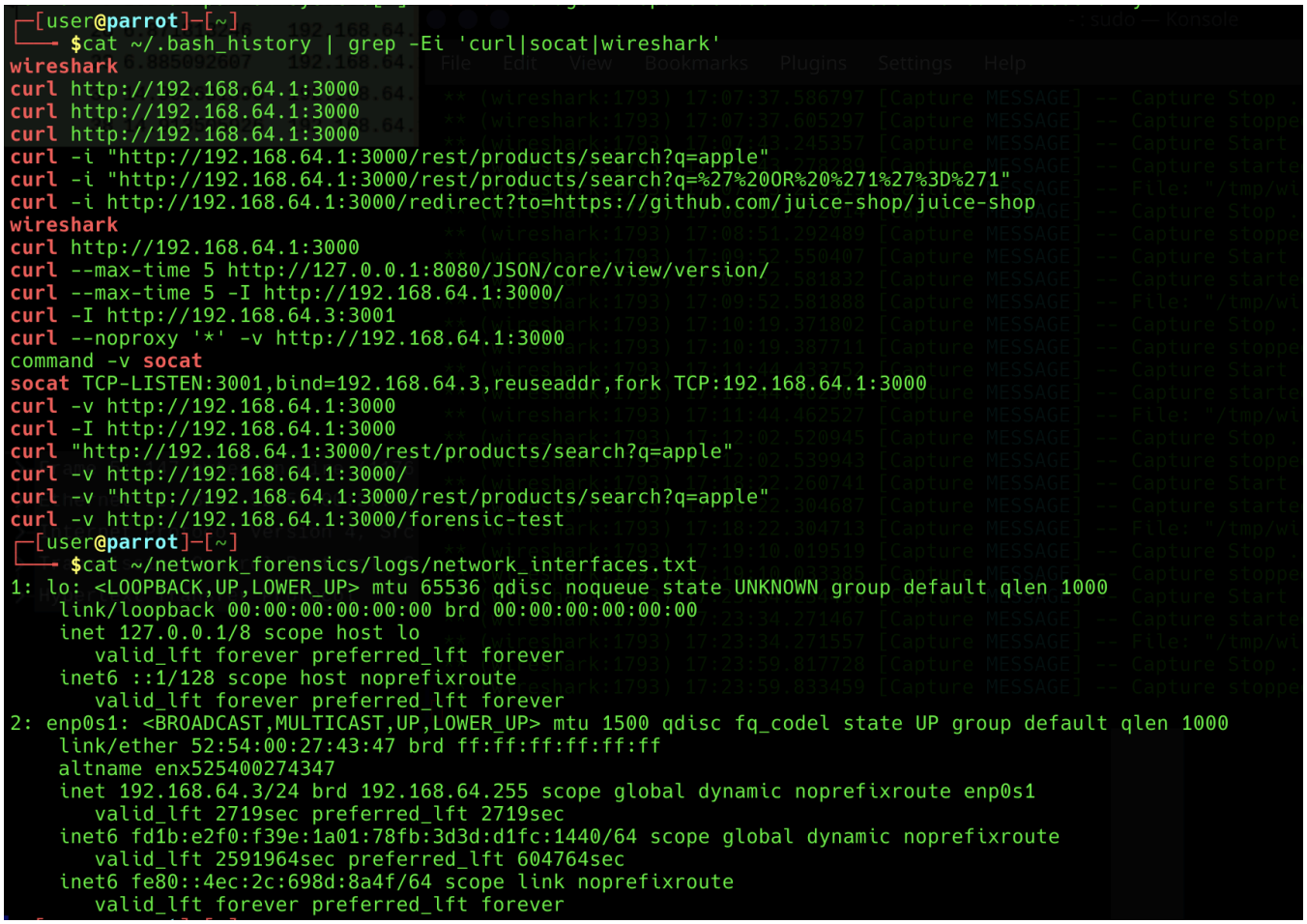
Microsoft Log Parser 2.2 provides a structured method for querying log data and identifying events relevant to a network forensic investigation. Its SQL-style filtering and aggregation capabilities can help reduce large log datasets and support correlation with packet-level evidence.

The collected Parrot OS logs and Wireshark capture provide the source evidence for this analysis. Completing and preserving the Log Parser query output will strengthen the documented evidence trail.

Appendix B — Screenshots

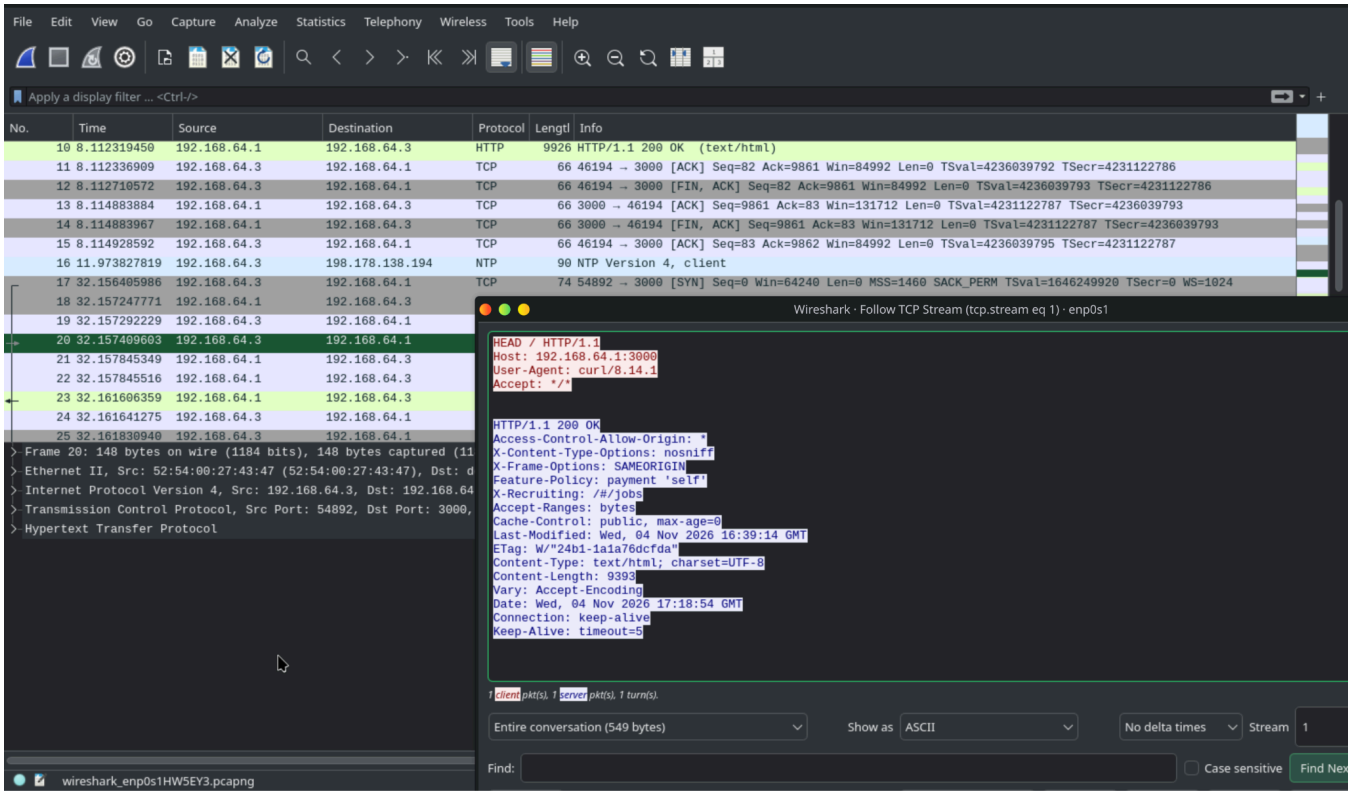
- 

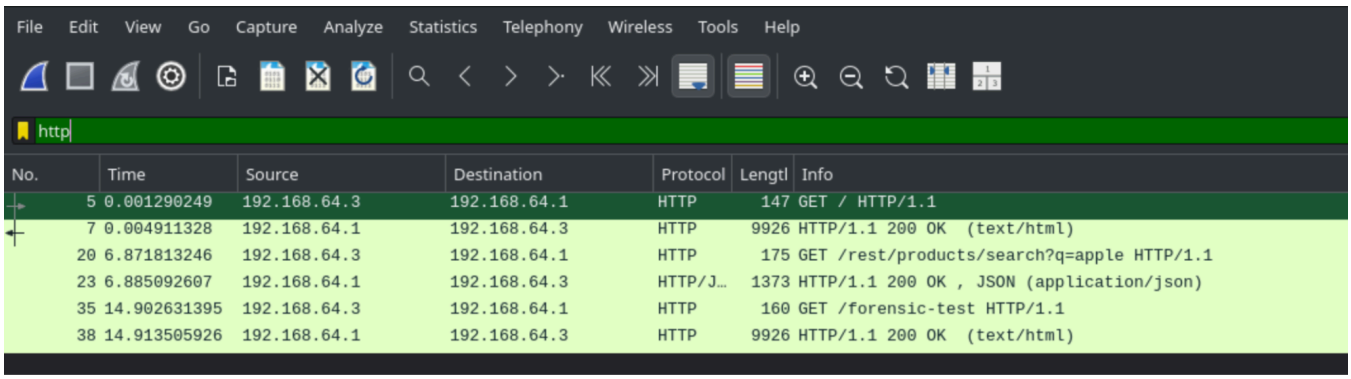
- 


┌─[user@parrot]─[~]
└─$cat ~/.bash_history | grep -Ei 'curl|socat|wireshark'
wireshark
curl http://192.168.64.1:3000
curl http://192.168.64.1:3000
curl http://192.168.64.1:3000
curl -i "http://192.168.64.1:3000/rest/products/search?q=apple"
curl -i "http://192.168.64.1:3000/rest/products/search?q=%27%20OR%20%271%27%3D%271"
curl -i http://192.168.64.1:3000/redirect?to=https://github.com/juice-shop/juice-shop
wireshark
curl http://192.168.64.1:3000
curl --max-time 5 http://127.0.0.1:8080/JSON/core/view/version/
curl --max-time 5 -I http://192.168.64.1:3000/
curl -I http://192.168.64.3:3001
curl --noproxy '*' -v http://192.168.64.1:3000
command -v socat
socat TCP-LISTEN:3001,bind=192.168.64.3,reuseaddr,fork TCP:192.168.64.1:3000
curl -v http://192.168.64.1:3000
curl -I http://192.168.64.1:3000
curl "http://192.168.64.1:3000/rest/products/search?q=apple"
curl -v http://192.168.64.1:3000/
curl -v "http://192.168.64.1:3000/rest/products/search?q=apple"
curl -v http://192.168.64.1:3000/forensic-test
┌─[user@parrot]─[~]
└─$cat ~/network_forensics/logs/network_interfaces.txt
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 52:54:00:27:43:47 brd ff:ff:ff:ff:ff:ff
altname enx525400274347
inet 192.168.64.3/24 brd 192.168.64.255 scope global dynamic noprefixroute enp0s1
valid_lft 2719sec preferred_lft 2719sec
inet6 fd1b:e2f0:f39e:1a01:78fb:3d3d:d1fc:1440/64 scope global dynamic noprefixroute
valid_lft 2591964sec preferred_lft 604764sec
inet6 fe80::4ec:2c:698d:8a4f/64 scope link noprefixroute
valid_lft forever preferred_lft forever

```
┌─[user@parrot]─[~]
└──╼ $cat ~/.bash_history | grep -Ei 'curl|socat|wireshark'
wireshark
curl http://192.168.64.1:3000
curl http://192.168.64.1:3000
curl http://192.168.64.1:3000
curl -i "http://192.168.64.1:3000/rest/products/search?q=apple"
curl -i "http://192.168.64.1:3000/rest/products/search?q=%27%20OR%20%271%27%3D%271"
curl -i http://192.168.64.1:3000/redirect?to=https://github.com/juice-shop/juice-shop
wireshark
curl http://192.168.64.1:3000
curl --max-time 5 http://127.0.0.1:8080/JSON/core/view/version/
curl --max-time 5 -I http://192.168.64.1:3000/
curl -I http://192.168.64.3:3001
curl --noproxy '*' -v http://192.168.64.1:3000
command -v socat
socat TCP-LISTEN:3001,bind=192.168.64.3,reuseaddr,fork TCP:192.168.64.1:3000
curl -v http://192.168.64.1:3000
curl -I http://192.168.64.1:3000
curl "http://192.168.64.1:3000/rest/products/search?q=apple"
curl -v http://192.168.64.1:3000/
curl -v "http://192.168.64.1:3000/rest/products/search?q=apple"
curl -v http://192.168.64.1:3000/forensic-test
┌─[user@parrot]─[~]
└──╼ $cat ~/network_forensics/logs/network_interfaces.txt
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 52:54:00:27:43:47 brd ff:ff:ff:ff:ff:ff
    altname enx525400274347
    inet 192.168.64.3/24 brd 192.168.64.255 scope global dynamic noprefixroute enp0s1
       valid_lft 2719sec preferred_lft 2719sec
    inet6 fd1b:e2f0:f39e:1a01:78fb:3d3d:d1fc:1440/64 scope global dynamic noprefixroute
       valid_lft 2591964sec preferred_lft 604764sec
    inet6 fe80::4ec:2c:698d:8a4f/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

```
┌─[user@parrot]─[~]
└──╼ $ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s1           UP             192.168.64.3/24 fd1b:e2f0:f39e:1a01:78fb:3d3d:d1fc:1440/64 fe80::4ec:2c:698d:8
a4f/64
┌─[user@parrot]─[~]
└──╼ $ip route
default via 192.168.64.1 dev enp0s1 proto dhcp src 192.168.64.3 metric 100
192.168.64.0/24 dev enp0s1 proto kernel scope link src 192.168.64.3 metric 100
┌─[user@parrot]─[~]
└──╼ $
```


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Lengtl Info
10 8.112319450 192.168.64.1 192.168.64.3 HTTP 9926 HTTP/1.1 200 OK (text/html)
11 8.112336909 192.168.64.3 192.168.64.1 TCP 66 46194 → 3000 [ACK] Seq=82 Ack=9861 Win=84992 Len=0 TSval=4236039792 TSecr=4231122786
12 8.112710572 192.168.64.3 192.168.64.1 TCP 66 46194 → 3000 [FIN, ACK] Seq=82 Ack=9861 Win=84992 Len=0 TSval=4236039793 TSecr=4231122786
13 8.114883884 192.168.64.1 192.168.64.3 TCP 66 3000 → 46194 [ACK] Seq=9861 Ack=83 Win=131712 Len=0 TSval=4231122787 TSecr=4236039793
14 8.114883967 192.168.64.1 192.168.64.3 TCP 66 3000 → 46194 [FIN, ACK] Seq=9861 Ack=83 Win=131712 Len=0 TSval=4231122787 TSecr=4236039793
15 8.114928592 192.168.64.3 192.168.64.1 TCP 66 46194 → 3000 [ACK] Seq=83 Ack=9862 Win=84992 Len=0 TSval=4236039795 TSecr=4231122787
16 11.973827819 192.168.64.3 198.178.138.194 NTP 90 NTP Version 4, client
17 32.156405986 192.168.64.3 192.168.64.1 TCP 74 54892 → 3000 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1646249920 TSecr=0 WS=1024
18 32.157247771 192.168.64.1 192.168.64.3
19 32.157292229 192.168.64.3 192.168.64.1
20 32.157409603 192.168.64.3 192.168.64.1
21 32.157845349 192.168.64.1 192.168.64.3
22 32.157845516 192.168.64.1 192.168.64.3
23 32.161606359 192.168.64.1 192.168.64.3
24 32.161641275 192.168.64.3 192.168.64.1
25 32.161830940 192.168.64.3 192.168.64.1
Frame 20: 148 bytes on wire (1184 bits), 148 bytes captured (11
Ethernet II, Src: 52:54:00:27:43:47 (52:54:00:27:43:47), Dst: d
Internet Protocol Version 4, Src: 192.168.64.3, Dst: 192.168.64
Transmission Control Protocol, Src Port: 54892, Dst Port: 3000,
Hypertext Transfer Protocol
wireshark_enp0s1HW5EY3.pcapng
Wireshark · Follow TCP Stream (tcp.stream eq 1) · enp0s1
HEAD / HTTP/1.1
Host: 192.168.64.1:3000
User-Agent: curl/8.14.1
Accept: */*
HTTP/1.1 200 OK
Access-Control-Allow-Origin: *
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Feature-Policy: payment 'self'
X-Recruiting: /#/jobs
Accept-Ranges: bytes
Cache-Control: public, max-age=0
Last-Modified: Wed, 04 Nov 2026 16:39:14 GMT
ETag: W/"24b1-1a1a76dcfda"
Content-Type: text/html; charset=UTF-8
Content-Length: 9393
Vary: Accept-Encoding
Date: Wed, 04 Nov 2026 17:18:54 GMT
Connection: keep-alive
Keep-Alive: timeout=5
1 client pkt(s), 1 server pkt(s), 1 turn(s).
Entire conversation (549 bytes)
Show as ASCII
No delta times
Stream 1
Find:
Case sensitive
Find Nex


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Source Destination Protocol Lengtl Info
5 0.001290249 192.168.64.3 192.168.64.1 HTTP 147 GET / HTTP/1.1
7 0.004911328 192.168.64.1 192.168.64.3 HTTP 9926 HTTP/1.1 200 OK (text/html)
20 6.871813246 192.168.64.3 192.168.64.1 HTTP 175 GET /rest/products/search?q=apple HTTP/1.1
23 6.885092607 192.168.64.1 192.168.64.3 HTTP/J… 1373 HTTP/1.1 200 OK , JSON (application/json)
35 14.902631395 192.168.64.3 192.168.64.1 HTTP 160 GET /forensic-test HTTP/1.1
38 14.913505926 192.168.64.1 192.168.64.3 HTTP 9926 HTTP/1.1 200 OK (text/html)